# LIB2019
# Water and Society


Dr Swatilekha Sen

Contact: swatilekha.sen@vitap.ac.in

# Urban Water Management

- Urban water management landscape of India
- Role of Geo-spatial technology in water resource management
- Marginal groups and watershed programmes in contemporary South India: The case study of Andhra Pradesh

## Urban water management landscape of India

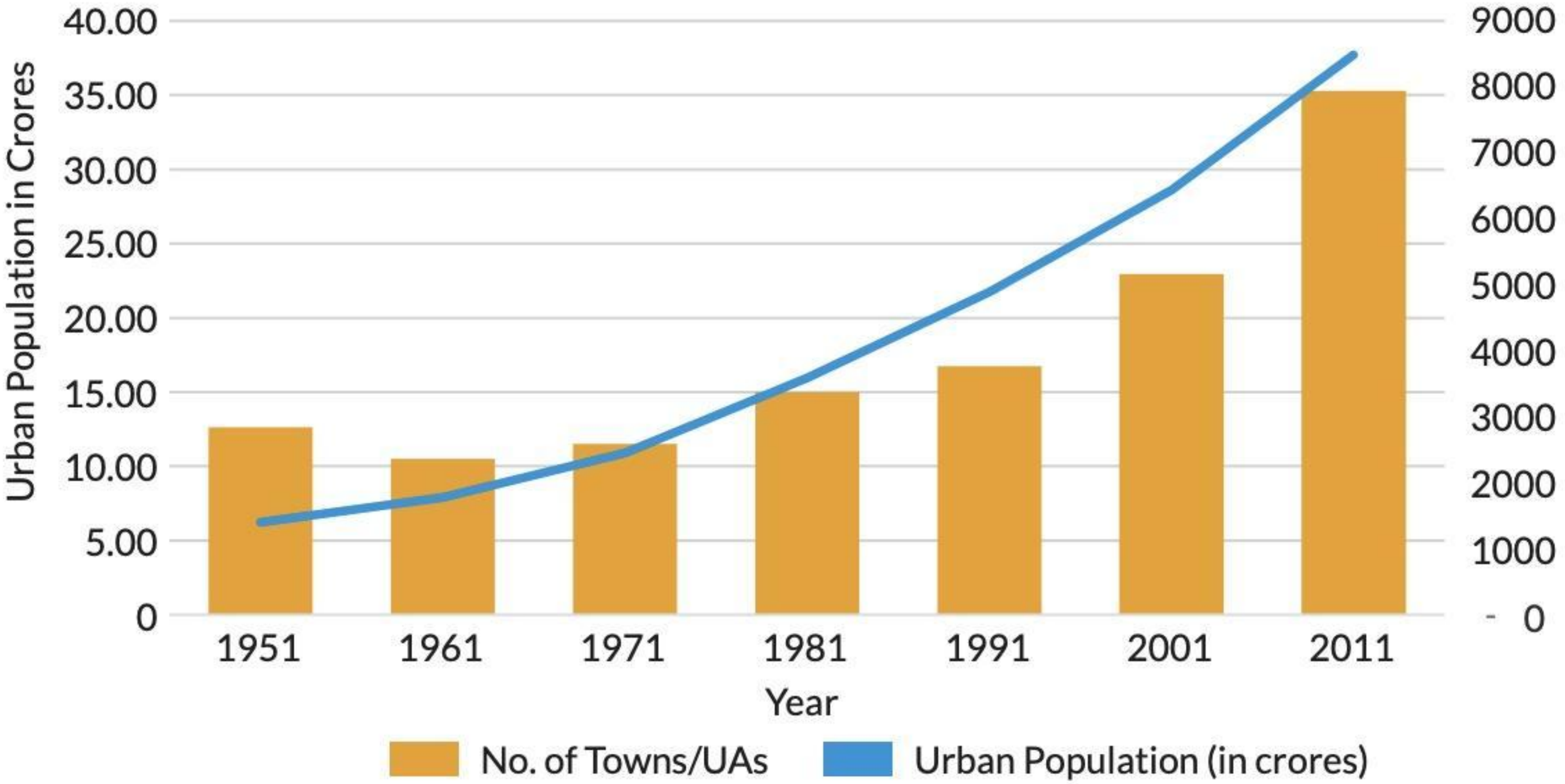
- Understanding the water supply sector in urban India, its linkages with health and climate, and providing aspects that need to be revisited to achieve the sanitation goals.
- As the world entered the 21st century, the Millennium Development Goals (MDG) were introduced to improve the lives of people.
- In 2015, the Sustainable Development Goals (SDG) were introduced to improve the lives and the environment across the globe, targeting holistic sustainable development.
- Out of the 17 SDGs covering various aspects of human habitats and life, SDG 6 focusses completely on water and sanitation. Other than this, water is an important factor that contributes to various other SDGs, directly or indirectly (Figure 1.1).

**Figure 1.1: The SDGs linked with Water management**


*Source: Author*

# Urbanisation and Urban Water Management in India

- The urban population has been constantly rising due to two factors- the natural growth of the urban areas, and the migration of people from rural areas to the urban areas.
- Such increasing population adds to the growing demands for resources, including water.
- Indian cities depend largely on groundwater within their boundaries for their freshwater needs, along with water abstracted from rivers sourced from distant reservoirs.
- This is despite the fact, that most cities have certain water bodies such as lakes and rivers within their geographic and administrative boundaries. However, most surface water sources in Indian cities are in a bad state, owing to pollution and contamination from domestic and/or industrial waste.




*Source- Adapted from Sadashivam & Tabassum (2016)*

# Stats about Urban Water at household level from census 2011 (CPHEEO, N.D.)

- 1. 62% of households depend on tapwater from treated sources 2. About 71% of households have drinking water source within their premises, while 8% of household don't have it.
- In India, the Twelfth Schedule provided by the Constitution (Seventy-fourth Amendment) Act, 1992 confers the Municipalities and Urban Local Bodies with the responsibility of providing the services and carrying out functions for providing water supply to the domestic, industrial and commercial consumers.
- *The per capita water supply levels are decided based on the city sanitation systems of the respective cities. The Central Public Health and Environmental Engineering Organisation (CPHEEO) has recommended the minimum water supply levels in the Manual on Water Supply and Treatment.

**Table 1.2: Recommended Water Supply Levels in Indian cities**

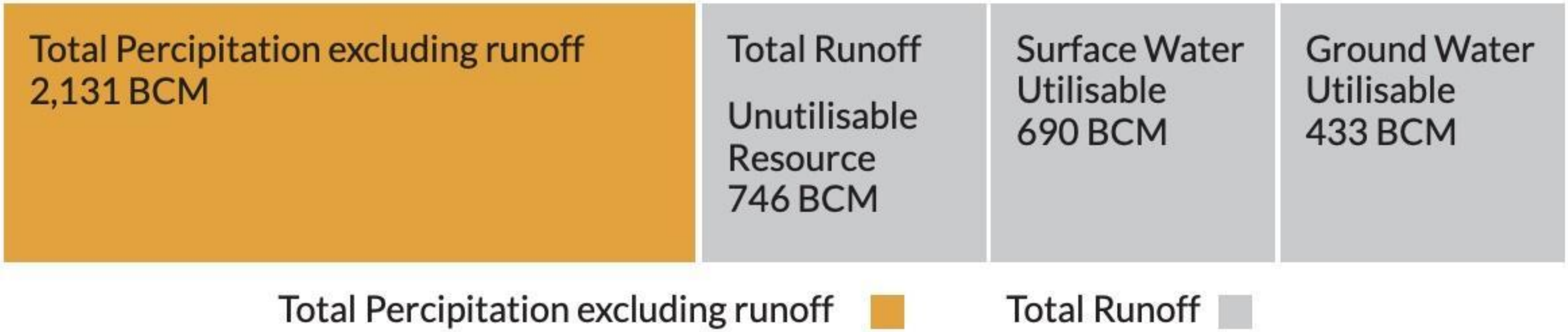
| Sr. No. | Classification of Towns/Cities | Recommended Minimum Water Supply Levels (lpcd) |
|---|---|---|
| 1 | Towns provided with piped water supply but without sewerage systems | 70 |
| 2 | Cities provided with piped water supply where sewerage system is existing/contemplated | 135 |
| 3 | Metropolitan and Mega Cities provided with piped water supply where sewerage system is existing/contemplated | 150 |

*Source – (CPHEEO, 1999)*

# Available Water Resources in India

- As of 2019, India is estimated to have about 17.8% of the world population residing in about 2.5% of the geographical area of the world (World Bank, 2020). In terms of water resources, India owns about 4% of the total water available in the world (ADRI, 2017). A total of 1869 Billion Cubic Metre (BCM) of water is the Mean Annual Natural Runoff flowing to the rivers and surface water bodies of India.
- However, only 1,123 BCM is the utilisable resource of water available, inclusive of both 433 BCM Ground Water and 690 BCM of Surface Water (CWC, 2016). The average annual per capita water availability in the years 2001 and 2011 was assessed as 1816 cubic meters and 1545 cubic meters respectively which may further reduce to 1486 cubic meters and 1367 cubic meters in the years 2021 and 2031 respectively (PIB, 2020).

**Figure 1.4: Available Water Resources in India**

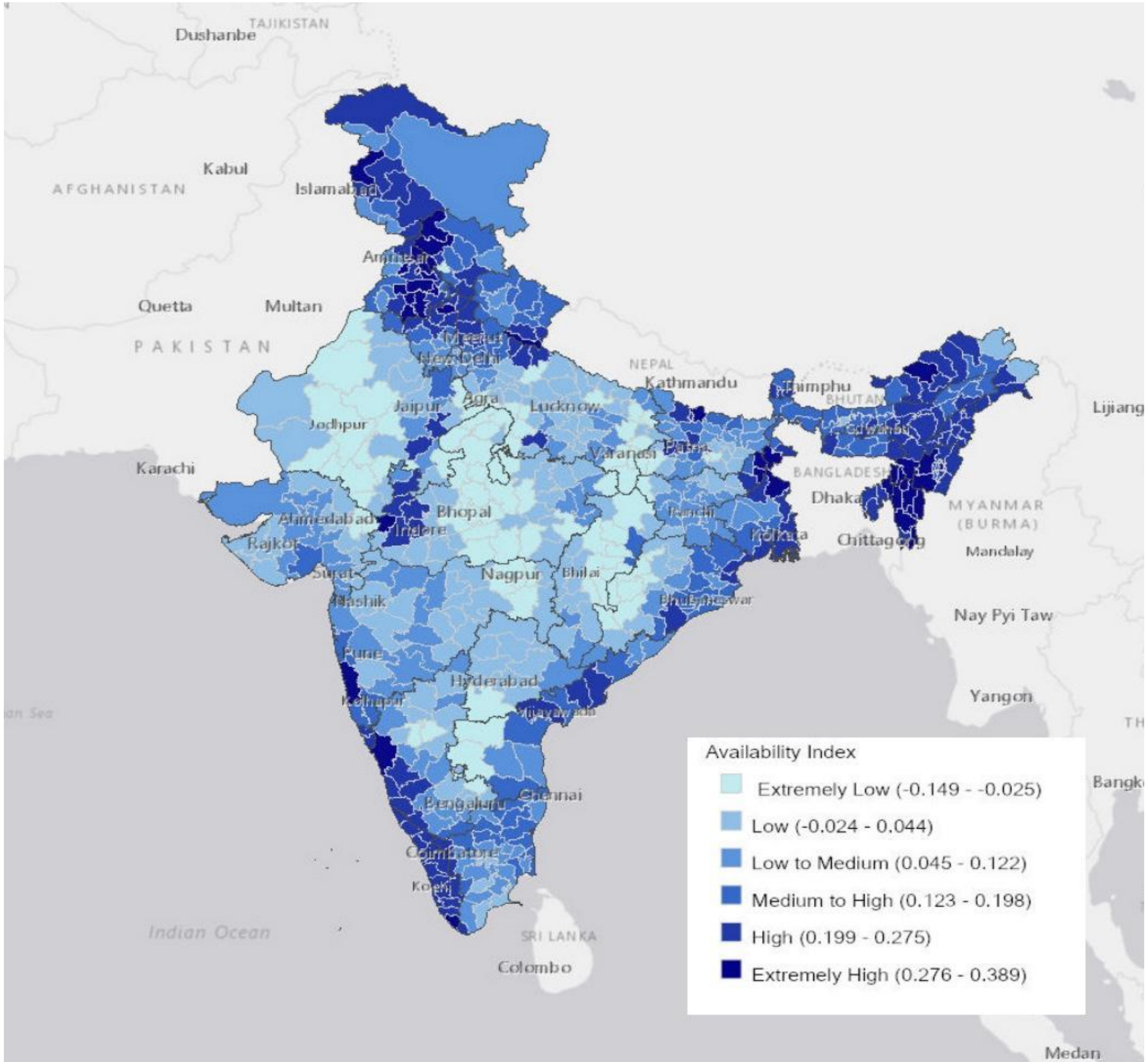



This water is distributed unequally across geographic regions, rural and urban areas, and social classes. Following the trend, India is water-stressed since 2011 and is anticipated to slide into the water-scarce zone by 2050 (Figure 1.5).

# Water in India- Demand, Supply, Stresses

- Water Availability Index is a measure that estimates surface water availability by recording water level in open surface water. Positive values are typically open water areas and negative values are non-water features (i.e. terrestrial vegetation, bare soil etc.).
- As per the calculations of the World Resources Institute (WRI) for the Water Availability Index, about two-thirds of India falls under extremely low to the medium category of the Index, denoting low available open surface water (Figure 1.6).
- However, this is staggeringly misleading if one compares population vis-à-vis the water available.

# Water Availability Index



# Legislative Framework in India for UWM

- Relevant Acts for water management in India

| Sr. No. | Act |
|---|---|
| 1 | The Water (Prevention and Control of Pollution) Cess Act, 1977 (GoI, 1977) |
| 2 | The Environment (Protection) Act, 1986 (GoI, 1984) |
| 3 | National Green Tribunal Act, 2010 (GoI, 2010) |

## The Water (Prevention and Control of Pollution) CESS Act, 1977 (GoI, 1977)

The Water CESS Act was passed to generate financial resources to meet expenses of the Central and State Pollution Boards. The Act creates economic incentives for pollution control and requires local authorities and certain designated industries to pay a CESS (tax) for water effluent discharge. The Central Government, after deducting the expenses of collection, pays the central and state boards such sums, as it seems necessary. To encourage capital investment in pollution control, the Act gives a polluter a 25% rebate of the applicable CESS upon installing effluent treatment equipment.

- The Environment (Protection) Act, 1986 (GoI, 1984)

This Act covers all forms of pollution; air, water, soil and noise. It provides the safe standards for the presence of various pollutants in the environment. It prohibits the use of hazardous material unless prior permission is taken from the Central Government and it allows the central government to assign authorities in various jurisdictions to carry out the laws of this Act.

- National Green Tribunal Act, 2010 (GoI, 2010)

The act provides for the establishment of a Green Tribunal, which is a special court dealing with cases related to environmental protection and conservation of natural resources and forests.

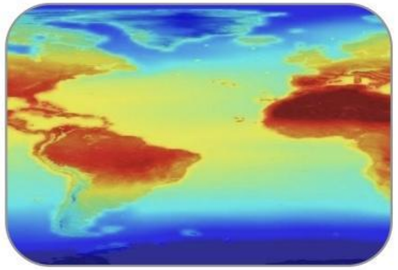
## Schemes and Programmes

# National Water Mission (National Water Mission, GoI, 2011)

- Comprehensive water data base in public domain and assessment of impact of climate change on water resource
- Promotion of citizen and state action for water conservation, augmentation and preservation
- Focused attention to vulnerable areas including over-exploited areas
- Increasing water use efficiency by 20%
- Promotion of basin level integrated water resource management

**Figure 2.2: Salient Features of National Water Mission**

Review of National Water Policy



Research and studies on all aspects related to impact of climate change on water resources including quality aspects of water resources

Expeditious implementation of water resources projects particularly the multipurpose projects with carry over storages

Promotion of traditional system of water conservation

Intensive programme for ground water recharge in over-exploited areas

Incentivize for recycling of water including wastewater

Planning on the principle of integrated water resources development and management

Ensuring convergence among various water resources programmes

Intensive capacity building and awareness programme including those for Panchayati Raj Institutions, urban local bodies and youths

# Atal Mission for Rejuvenation and Urban Transformation (AMRUT) (MoHUA, GoI, 2015)

**Water supply:**
Augmentation of existing water supply, water treatment plants and universal metering.
Rehabilitation of old water supply systems, including treatment plants.
Rejuvenation of water bodies specially for drinking water supply and recharging of ground water.
Special water supply arrangement for difficult areas, hill and coastal cities, including those having water quality problems (e.g. arsenic, fluoride).
Recycling/reuse of water water and reduction of non-revenue water.

Sewarage facilities and septage management

Storm water drains to reduce flooding

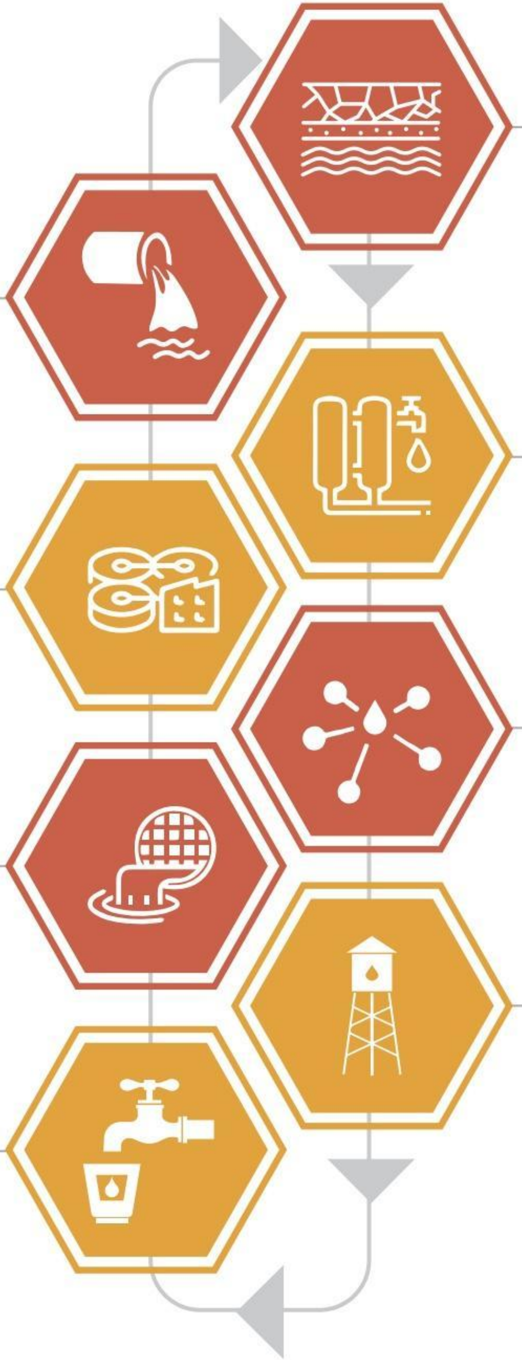
Pedestrian, non-motorized and public transport facilities, parking spaces

Enhancing amenity value of cities by creating and upgrading green spaces, park and recreation centers, especially for children

# Urban Water Cycle

- The basics of the hydrological cycle, or the water cycle, are condensation, precipitation, transportation, and evaporation. These processes operate on global scales and in natural environments. But on local scales and in engineered environments like cities, a different cycle dominates — the urban water cycle.
- The urban water cycle includes components such as water treatment, distribution, stormwater drainage, wastewater drainage, wastewater treatment, recycling or disposal. This consists of an interaction of the natural water cycle with the social water cycle.

**Figure 3.1: Stages of Urban Water Cycle**



**Source**
The water is extracted from various sources like ground water, rivers etc. to meet the requirements of the population.

**Treatment**
Water extracted from the source which can have pathogens, pollutants or other harmful substances, will undergo treatment processes to make it suitable for drinking and other house hold uses.

**Distribution**
After treatment and sufficient disinfection, the water is sent out via a pressurized system of lifts to the areas in the city where it is needed. A disinfectant residual must be maintained at all parts of the system.

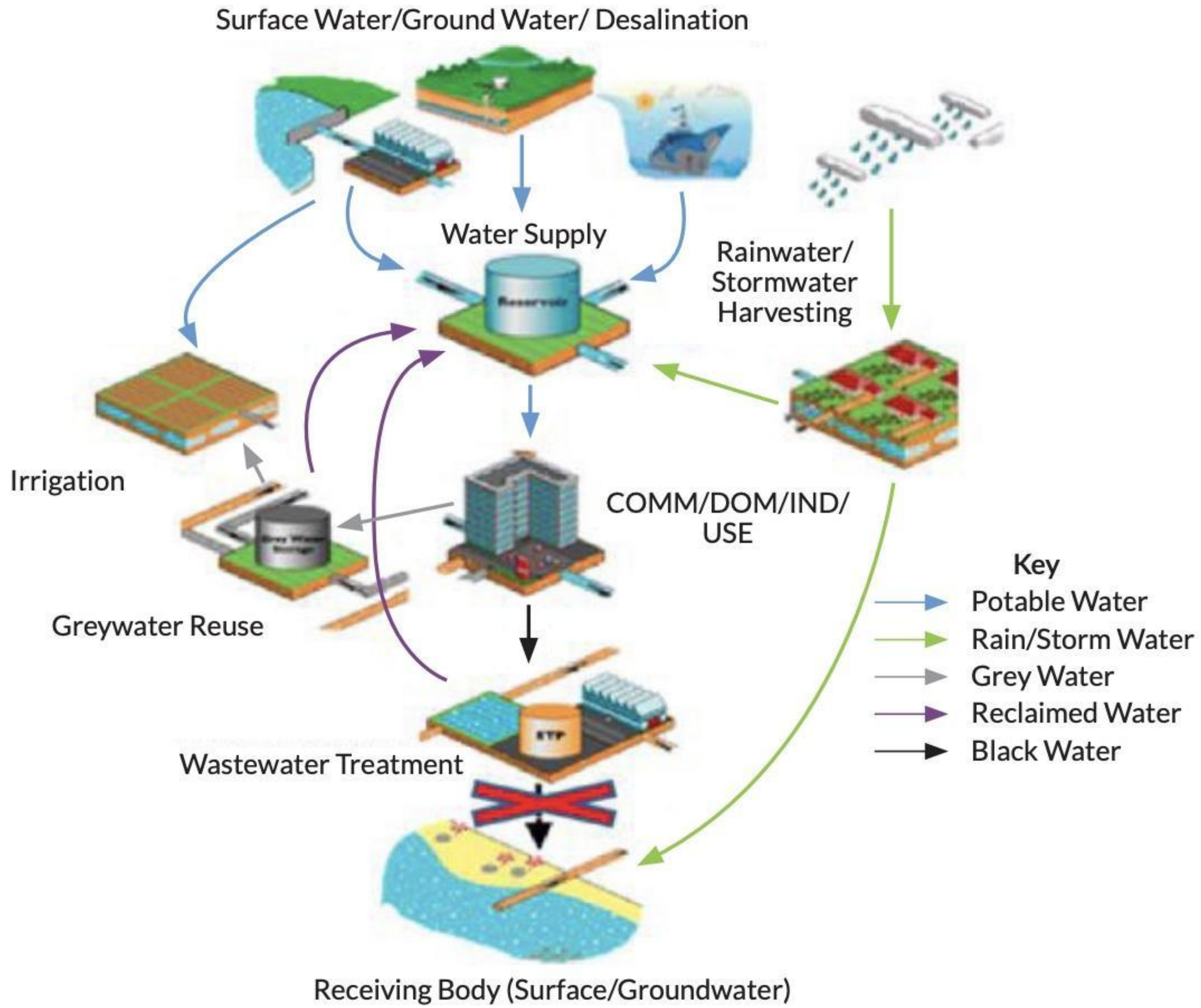
**Storage**
After being distributed, the water is stored in water towers before it is used. Water towers use gravity to ensure uninterrupted supply.

**Use**
Water is used for various purposes in households varying from drinking and cooking to washing, gardening and flushing toilets.

**Collection**
Water leaves our homes by going down our drains and toilets. Sewer systems collect and convey the wastewater from our homes and businesses to wastewater treatment plants. The process is typically done using gravity.

**Treatment**
Because the water quality is reduced by usage, and because it has to be put back into the hydrologic cycle, it has to be treated. This is done through a variety of biological and chemical processes.

**Discharge**
Once the treated water (effluent) has been cleaned to regulatory standards, it is discharged back into the environment. When it gets back to the environment, the cycle starts again.




*Source — Khatri, et al. (2014)*

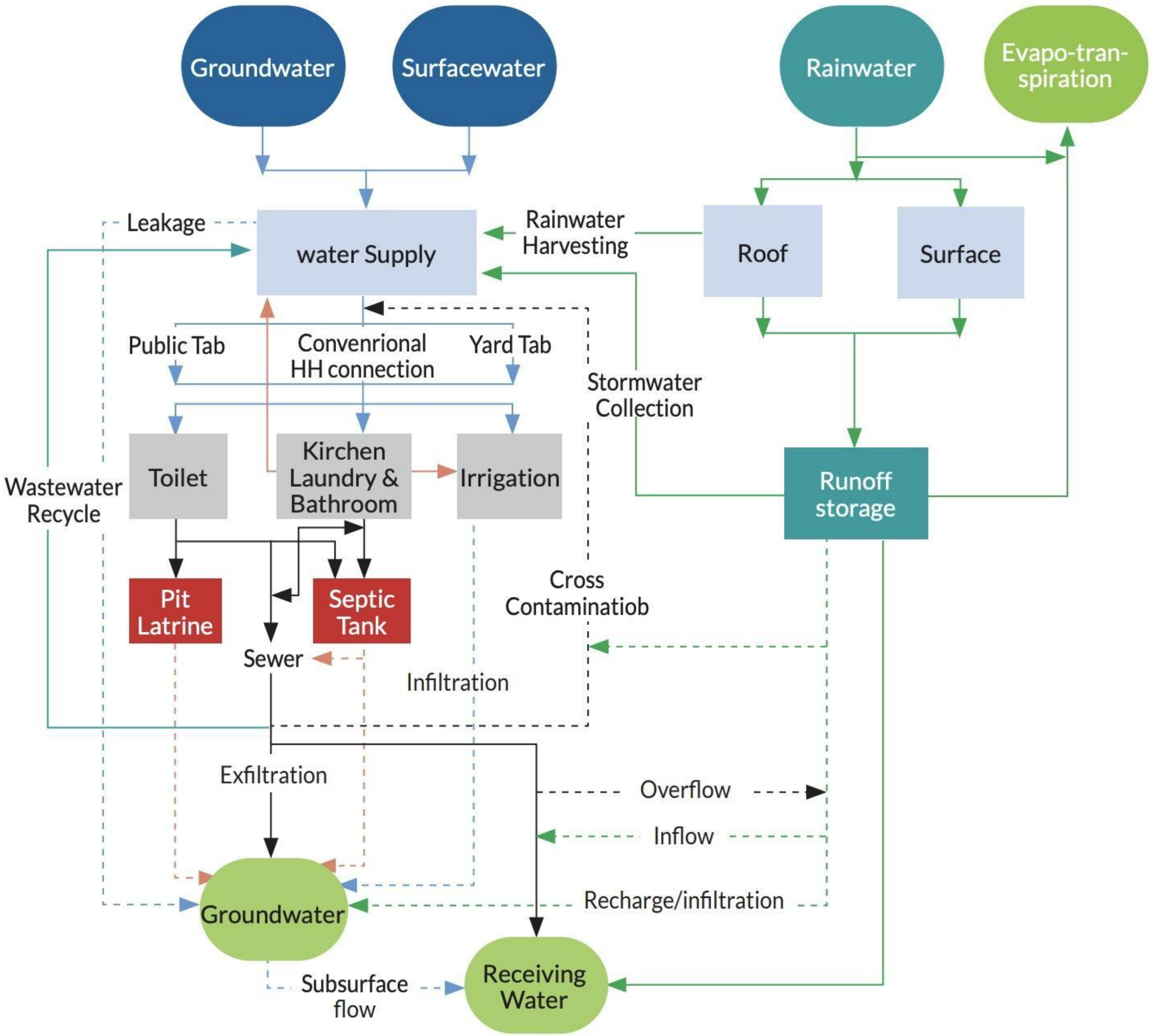
- The excessive human intervention caused some distortions to the natural solar water cycle. A large area of impervious pavements affects the groundwater recharging and result in an increased run-off to the sea.
- To tackle this, direct extraction of precipitation and artificial recharging of groundwater has been practised in various cities.
- Climate change poses a great threat to the existing urban water cycle; there is a noticeable increase in the variability of precipitation and the reliability of water sources throughout the year.
- Efficient water management strategies are thus required for ensuring the sustainability of water services in cities, because water unlike most resources, does not stay limited to any geopolitical and physical boundaries created by man.
- Because of this transient and fluid nature of water, any pressure or impact to the water resources or the dynamics of the interaction of water with other resources, such as soil, air, climate, etc., results in changes in the dynamics of the water cycle.

# Concept of Integrated Urban Water Management

- Integrated Urban Water Management (IUWM) is an approach that seeks to develop efficient and flexible urban water systems by adopting a diversity of existing technologies, management, and institutional practices to supply and secure water for urban areas.
- The characteristics of IUWM are:

- Recognizes alternative water sources.
- Differentiates the qualities and potential uses of water sources.
- Consideration of water storage, distribution, treatment, recycling, and disposal as part of the same resource management cycle.
- Seeks to protect, conserve and exploit water at its source.
- Accounts for rural and peri-urban users that are dependent on the same water source.
- Aligns formal institutions (organizations, legislation, and policies) and informal practices (norms and conventions) that govern water in and for cities.
- Recognizes the relationships between water resources, land use, and energy.
- Simultaneously pursues economic efficiency, social equity, and environmental sustainability.
- Encourages participation by all stakeholders.

**Figure 3.3: Integrated Urban Water Management framework for Low-Income Areas**




*Source — Adapted from Khatri, et al. (2014)*

**Figure 4.1: Water Sensitive Urban Design**

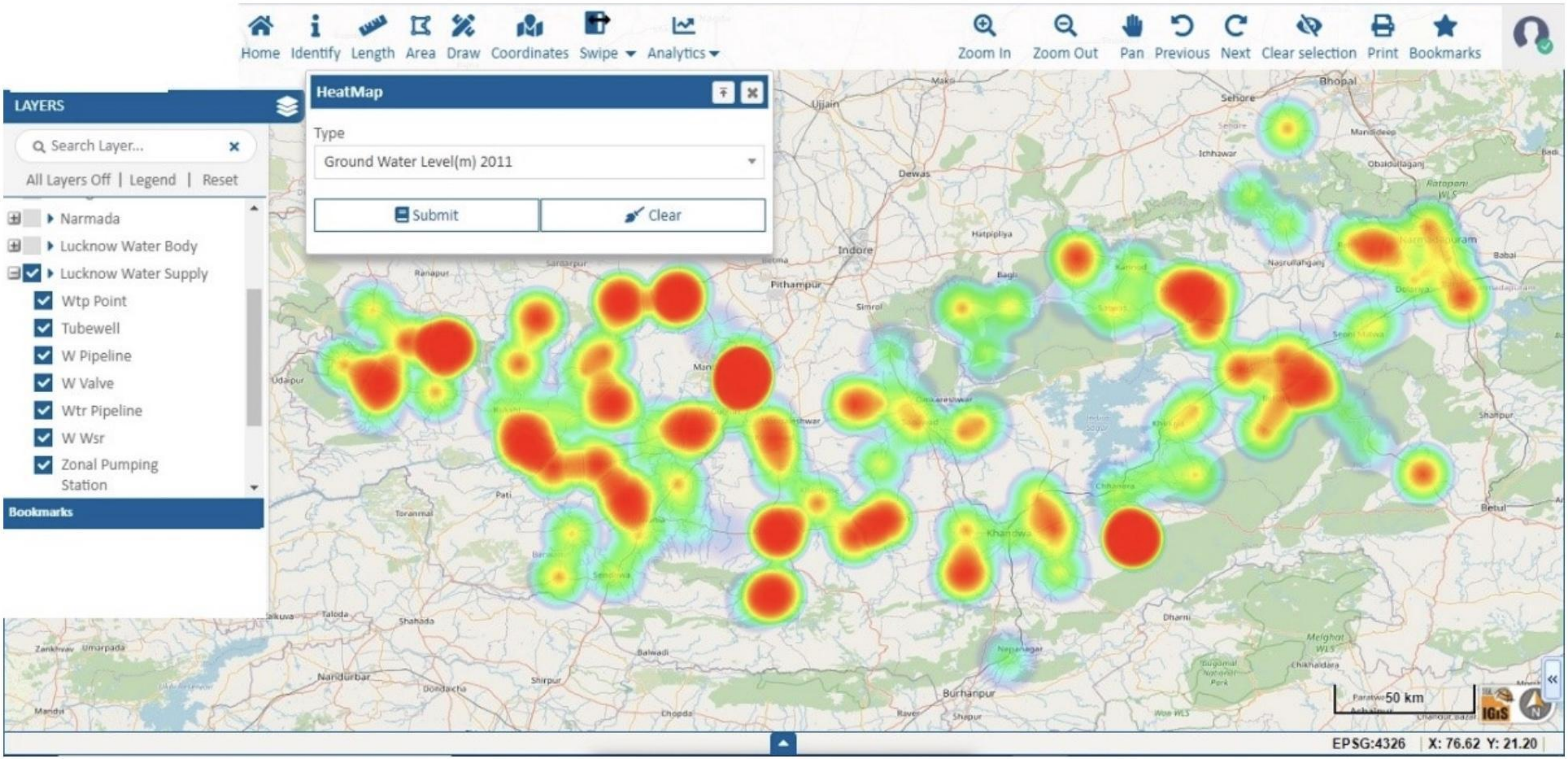

*Source — Morgan, et al. (2013)*

# Role of Geo-Spatial Technology (GIS) in Water Resource Management

- Current research on water availability in India focuses on identifying areas of water stress and developing strategies to manage water resources more effectively.
- The implementation of advanced geospatial technology has made water resource management activities easier. It includes various water-related data collection, data storage, management, and analysis.
- IGiS, an indigenous technology, jointly developed by SGL and SAC, ISRO, provides valuable insights for efficient decision-making activities and helps to implement better water management policies through various geospatial analyses for sustainable water resource management.
- These are beneficial for mapping the water resources, and assets, monitoring water availability, and water quality, identifying potential zones of groundwater, water supply system management, and analyzing the impact of human activities on water resources effectively and efficiently.

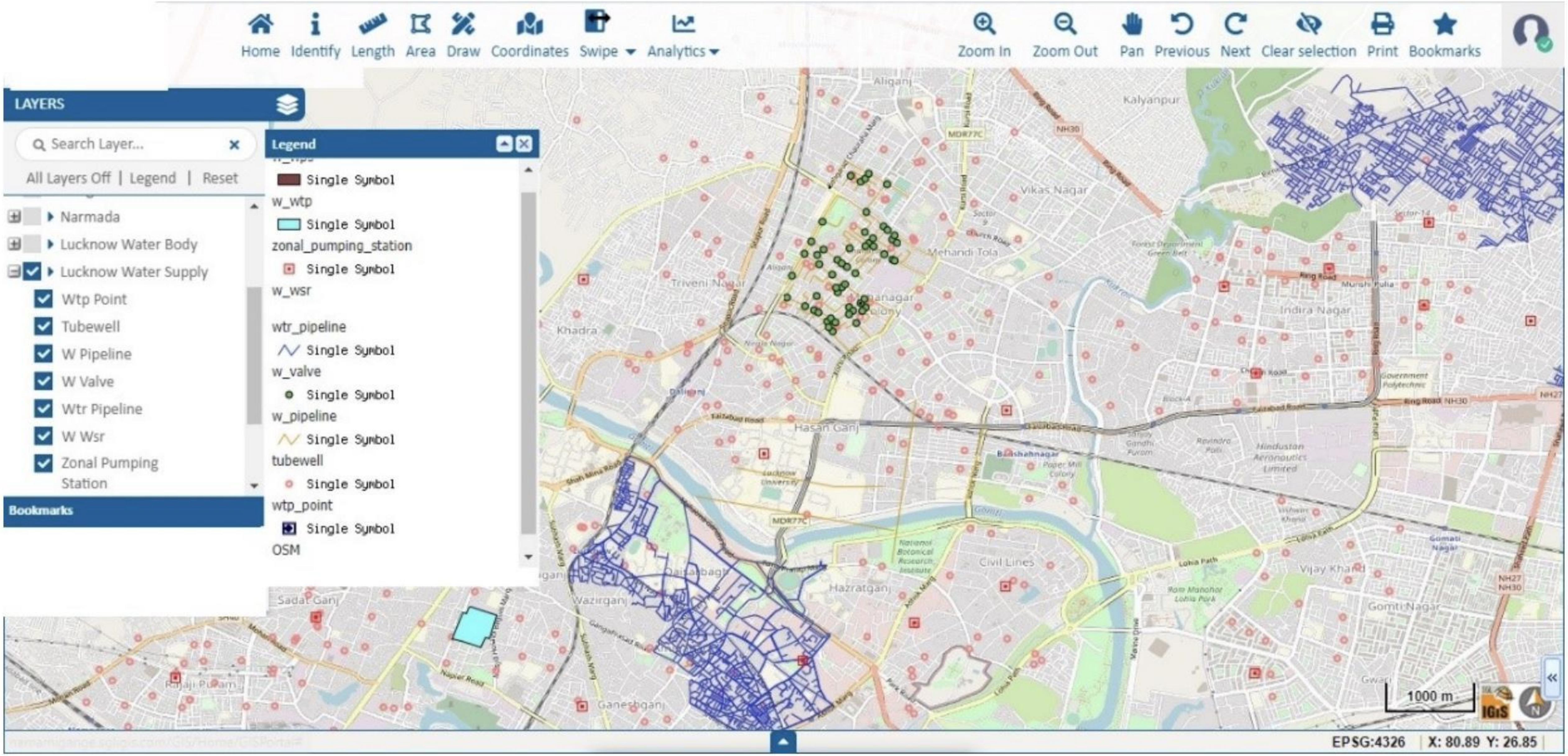
## Ground Water Management

- GIS can be used to monitor and inspect water resources through comprehensive maps of groundwater resources, taking into account aquifer boundaries, water levels, and quality information which comes from remotely sensed images and sensors installed in the field. The platform incorporates diverse image-processing techniques that facilitate the identification of potential groundwater zones. It also enables the solution for how groundwater levels will respond when there are changes in rainfall amounts, pumping rates, or other factors. Thus, the integration of GIS technology in the groundwater management system, enhances decision-making, promotes conservation efforts, and facilitates sustainable use of this vital resource.


Home
Identify
Length
Area
Draw
Coordinates
Swipe
Analytics
Zoom In
Zoom Out
Pan
Previous
Next
Clear selection
Print
Bookmarks
LAYERS
Search Layer...
All Layers Off | Legend | Reset
Narmada
Lucknow Water Body
Lucknow Water Supply
Wtp Point
Tubewell
W Pipeline
W Valve
Wtr Pipeline
W Wsr
Zonal Pumping Station
Bookmarks
HeatMap
Type
Ground Water Level(m) 2011
Submit
Clear
Ujjain
Bhopal
Sehore
Dewas
Indore
Pithampur
Khandwa
Burhanpur
Nandurbar
Betul
50 km
EPSG:4326
X: 76.62 Y: 21.20

## Water Supply Management

- This includes creating detailed maps of water treatment plants, reservoirs, pipelines, and distribution networks, assessing population density, and other useful data. With GIS, authorities can monitor the performance of the entire water system, identify areas with high demand or potential leakage, and decide on the best locations for new infrastructure. Additionally, GIS can be used to analyse water supply scenarios like the demand for water in a particular area for giving insight to develop effective strategies for managing water sources. Ultimately, GIS plays a major role in ensuring the sustainable management of water resources.


Home
Identify
Length
Area
Draw
Coordinates
Swipe
Analytics
Zoom In
Zoom Out
Pan
Previous
Next
Clear selection
Print
Bookmarks
LAYERS
Search Layer...
All Layers Off | Legend | Reset
Narmada
Lucknow Water Body
Lucknow Water Supply
Wtp Point
Tubewell
W Pipeline
W Valve
Wtr Pipeline
W Wsr
Zonal Pumping Station
Bookmarks
Legend
Single Symbol
w_wtp
Single Symbol
zonal_pumping_station
Single Symbol
w_wsr
wtr_pipeline
Single Symbol
w_valve
Single Symbol
w_pipeline
Single Symbol
tubewell
Single Symbol
wtp_point
Single Symbol
OSM
Aliganj
Kalyanpur
Vikas Nagar
Mehandi Tola
Triveni Nagar
Khadra
Indira Nagar
Hasan Ganj
Civil Lines
Hazratganj
Vijay Khand
Gomti Nagar
Sadat Ganj
Wazirganj
Ganeshganj
Rajaji Puram
1000 m
EPSG:4326
X: 80.89 Y: 26.85

## Water Quality Monitoring:

- GIS-based water quality monitoring involves the real-time quality monitoring of various water bodies, such as rivers, lakes, reservoirs, etc. It helps in understanding the spatial distribution of water quality parameters, identifying pollution sources, and implementing effective management strategies. Real-time water quality monitoring is required to ensure that all humankind and living creatures use safe water in their day-to-day life. Therefore, Internet of Things (IoT) based monitoring systems are used for constant monitoring of the water quality.
- Data collected from the IoT Sensors integrate with the GIS-based system for continuous monitoring of the information over the map. The GIS system is also capable of generating alarms when any abnormal changes are found. This information is vital for decision-making related to water quality management, such as selecting appropriate pollution control measures, identifying priority areas for water quality improvement, and assessing the effectiveness of water quality management programs.

# Infrastructure Management:

- Implementation of GIS in Infrastructure Management can be used to model the impact of population growth and other factors on water demand and infrastructure needs. GIS allows for accurate mapping and management of water supply and wastewater infrastructure, including pipes, treatment plants, reservoirs, and distribution networks.
- By digitizing and visualizing this infrastructure data, it becomes easier to identify areas of concern, detect leaks, optimize maintenance schedules, and plan for necessary repairs or upgrades. IGiS also allows the integration of real-time data from sensors and monitoring devices, enhancing the ability to respond promptly to any issues or disruptions in the water supply.
- This valuable information aids in optimizing the operation and maintenance of these systems, minimizing leaks and water losses, and strategically planning for future infrastructure demands.
- Moreover, the Multi-criterion decision-making tool in GIS allows for a comprehensive assessment of different criteria or factors involved in the site selection process such as rainfall patterns, topography, soil type, land use, and proximity to water sources. By considering multiple criteria simultaneously, the tool helps in evaluating and ranking potential sites based on their suitability for Rain Water Harvesting.